

This Zenodo deposit contains a publicly available description of the Dataset:

"Single-cell RNAseq of human PBMCs from healthy control, RBD, and PD."

Dataset Description:

We performed 10X Genomics single-cell RNAsequencing of human peripheral blood mononuclear cells from healthy control, PD and RBD patients. This dataset contains raw FASTQ files. Sequencing was performed using NovaSeq 6000 S4 PE 100bp. Reads were processed using the 10X Genomics Cell Ranger Single Cell 2.0.0 pipeline. FASTQs generated from sequencing output were aligned to the human GRCh38 reference genome using STAR algorithm 2.7.3a.

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Dataset Name: desjardins-human-pbmc-multimodal-sc-rna-tcr, v1.0

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ASAP Lab:

ASAP Project: Constructing a single cell atlas of human PBMCs from healthy control, RBD, and PD.

Project Description: Parkinson's disease (PD) is a neurodegenerative disorder marked by the development of cardinal motor deficits preceded by a protracted prodromal period of non-motor symptoms, which may include REM-sleep behavioural disorder (RBD). There is an emerging consensus that both the peripheral immune system and local neuroinflammation play key roles in the etiology of PD. Current work leaves many questions regarding the role of the immune system in the etiology of PD. This work aims to construct a comprehensive atlas comparing the transcriptome of HC, RBD, and PD across diverse PBMC cell subtypes.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset